

Unit-2 : COMBINATORICS

Mathematical induction - The pigeonhole principle - permutation and combinations - Recurrence relations - solving linear recurrence relations - Inclusion and exclusion principle and its applications.

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PART-B

1. Find the number of positive integers ≤ 1000 and not divisible any number of 3, 5, 7, 22?

SOLUTION:

let A denote integers ≤ 1000 that are divisible by 3

let B denote integers ≤ 1000 that are divisible by 5

let C denote integers ≤ 1000 that are divisible by 7

let D denote integers ≤ 1000 that are divisible by 22

$$[A] = \left\lfloor \frac{1000}{3} \right\rfloor = 333, [B] = \left\lfloor \frac{1000}{5} \right\rfloor = 200, [C] = \left\lfloor \frac{1000}{7} \right\rfloor = 142$$

$$[D] = \left\lfloor \frac{1000}{22} \right\rfloor = 45$$

$$\text{Now, } [A \cap B] = \left\lfloor \frac{1000}{3 \times 5} \right\rfloor = 66, [A \cap C] = \left\lfloor \frac{1000}{3 \times 7} \right\rfloor = 47$$

$$[A \cap D] = \left\lfloor \frac{1000}{3 \times 22} \right\rfloor = 15, [B \cap C] = \left\lfloor \frac{1000}{5 \times 7} \right\rfloor = 28$$

$$[B \cap D] = \left\lfloor \frac{1000}{5 \times 22} \right\rfloor = 9, [C \cap D] = \left\lfloor \frac{1000}{7 \times 22} \right\rfloor = 6$$

$$[A \cap B \cap C] = \left\lfloor \frac{1000}{3 \times 5 \times 7} \right\rfloor = 9, [A \cap B \cap D] = \left\lfloor \frac{1000}{3 \times 5 \times 22} \right\rfloor = 3$$

$$[A \cap C \cap D] = \left\lfloor \frac{1000}{3 \times 7 \times 22} \right\rfloor = 2, [B \cap C \cap D] = \left\lfloor \frac{1000}{5 \times 7 \times 22} \right\rfloor = 1$$

$$[A \cap B \cap C \cap D] = \left\lfloor \frac{1000}{3 \times 5 \times 7 \times 22} \right\rfloor = 0.$$

The number of integers ≤ 1000 , that are divisible by 3, 5, 7 and 22 is

$$[A \cup B \cup C \cup D]$$

By principle of Inclusion and Exclusion

$$\begin{aligned} [A \cup B \cup C \cup D] &= [A] + [B] + [C] + [D] \\ &\quad - [A \cap B] - [A \cap C] - [A \cap D] - [B \cap C] \\ &\quad - [B \cap D] - [C \cap D] + [A \cap B \cap C] + [A \cap B \cap D] \\ &\quad + [A \cap C \cap D] + [B \cap C \cap D] - [A \cap B \cap C \cap D] \end{aligned}$$

$$\begin{aligned} (\text{i.e.}) [A \cup B \cup C \cup D] &= (333 + 200 + 142 + 45) \\ &\quad - (66 + 47 + 15 + 28 + 9 + 6) \\ &\quad + (9 + 3 + 2 + 1) - 0 \end{aligned}$$

$$[A \cup B \cup C \cup D] = (720 - 171 + 15) = 564$$

The no. of integers not divisible by any of 3, 5, 7 and 22

$$= \text{TOTAL} - [A \cup B \cup C \cup D]$$

$$= 1000 - 564 = 436$$

Q (i) Find the number of integers between 1 and 500 that are divisible by integers 2, 3 and 5.

(ii) show that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$ by mathematical Induction method.

SOLUTION:

(i) Let A, B and C denote respectively the number of integer between 1 to 500 that are divisible by 2, 3 and 5 respectively

$$[A] = \left\lfloor \frac{500}{2} \right\rfloor = 250, \quad [B] = \left\lfloor \frac{500}{3} \right\rfloor = 166,$$

$$[C] = \left\lfloor \frac{500}{5} \right\rfloor = 100.$$

$$[A \cap B] = \left\lfloor \frac{500}{2 \times 3} \right\rfloor = 83, \quad [A \cap C] = \left\lfloor \frac{500}{2 \times 5} \right\rfloor = 50$$

$$[B \cap C] = \left\lfloor \frac{500}{3 \times 5} \right\rfloor = 33.$$

$$[A \cap B \cap C] = \left\lfloor \frac{500}{2 \times 3 \times 5} \right\rfloor = 16$$

The number of integers between 1 to 500 that are divisible by 2, 3 and 5

By the principle of inclusion and exclusion

$$[A \cup B \cup C] = [A] + [B] + [C] - [A \cap B] - [A \cap C] - [B \cap C] + [A \cap B \cap C]$$

$$[A \cup B \cup C] = 250 + 166 + 100 - 83 - 50 - 33 + 16$$

$$\text{i.e., } [A \cup B \cup C] = 366$$

(11)

$$\text{let } P(n) = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Step 1.

$$\text{put } n=1$$

$$P(1) = \frac{1(1+1)(2+1)}{6} = \frac{6}{6} = 1 = 1^2$$

$$\text{i.e., } \boxed{P(1) = 1^2}$$

so $P(1)$ is true

Step 2.

$$\text{put } n=k$$

$$P(k) = 1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

so $P(k)$ is true

Step 3.

$$\text{put } n=k+1$$

$$P(k+1) = (1^2 + 2^2 + 3^2 + \dots + k^2) + (k+1)^2$$

$$P(k+1) = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$\text{(i.e.,)} \quad P(k+1) = \frac{k(k+1)(2k+1) + 6(k+1)^2}{6}$$

$$P(k+1) = \frac{(k+1)}{6} \{ k(2k+1) + 6(k+1) \}$$

$$P(k+1) = \frac{(k+1)}{6} (2k^2 + k + 6k + 6)$$

$$P(K+1) = \frac{(K+1)}{6} (2K^2 + 7K + 6)$$

$$P(K+1) = \frac{(K+1)}{6} (2K^2 + 4K + 3K + 6)$$

$$\begin{pmatrix} 12 \\ 4 \times 3 \\ 7 \end{pmatrix}$$

$$P(K+1) = \frac{(K+1)}{6} (2K(K+2) + 3(K+2))$$

$$P(K+1) = \frac{(K+1)}{6} (K+2) (2K+3)$$

$$P(K+1) = \frac{1}{6} (K+1) ((K+1)+1) (2(K+1)+1)$$

$\therefore P(K+1)$ is true whenever $P(K)$ is true

By the principle of mathematical Induction $P(n)$ is true for all positive Integer n .

3. (i) Find the number of integers ≤ 500 that are not divisible by integers 2, 3 and 5.
 (ii) solve the recurrence relation for Fibonacci series.

SOLUTION:

(i) Let A , B and C denote respectively the number of Integer between 1 to 500 that are divisible by 2, 3 and 5 respectively.

$$[A] = \left\lfloor \frac{500}{2} \right\rfloor = 250, [B] = \left\lfloor \frac{500}{3} \right\rfloor = 166$$

$$[C] = \left\lfloor \frac{500}{5} \right\rfloor = 100, [A \cap B] = \left\lfloor \frac{500}{2 \times 3} \right\rfloor = 83$$

$$[A \cap C] = \left\lfloor \frac{500}{2 \times 5} \right\rfloor = 50, [B \cap C] = \left\lfloor \frac{500}{3 \times 5} \right\rfloor = 33$$

$$[A \cap B \cap C] = \left\lfloor \frac{500}{2 \times 3 \times 5} \right\rfloor = 16$$

The number of integers between 1 to 500 that are divisible by 2, 3 and 5. By the principle of Inclusion and Exclusion.

$$[A \cup B \cup C] = [A] + [B] + [C] - [A \cap B] - [A \cap C] - [B \cap C] + [A \cap B \cap C]$$

$$(i.e.) [A \cup B \cup C] = 250 + 166 + 100 - 83 - 50 - 33 + 16 = 366$$

$\therefore$ The number of integers not divisible by any of 2, 3 and 5

$$= \text{TOTAL} - [A \cup B \cup C] = 500 - 366$$

$$= 134$$

(ii) Fibonacci sequence $0, 1, 1, 2, 3, 5, 8, \dots$ satisfies the recurrence relation

$$f_n = f_{n-1} + f_{n-2}$$

$\Rightarrow \boxed{f_n - f_{n-1} - f_{n-2} = 0}$ and also satisfies the initial conditions are $f_0 = 0$ and $f_1 = 1$.

Given, $f_n - f_{n-1} - f_{n-2} = 0$

let $\boxed{f_n = r^n}$

(i.e.) $r^n - r^{n-1} - r^{n-2} = 0$

$$\Rightarrow r^n (1 - r^{-1} - r^{-2}) = 0$$

$$\Rightarrow 1 - r^{-1} - r^{-2} = 0 \Rightarrow 1 - \frac{1}{r} - \frac{1}{r^2} = 0$$

$$\Rightarrow \boxed{r^2 - r - 1 = 0} \quad \text{solving}$$

$$r = \frac{1 \pm \sqrt{1+4}}{2 \times 1} = \frac{1 \pm \sqrt{5}}{2}$$

Solution $f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)^n \rightarrow \textcircled{A}$

To find: α_1 & α_2 with initial condition

$$f_0 = 0 \text{ \& \> } f_1 = 1$$

put $n=0$ in eqn (A)

$$f_0 = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right)^0 + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)^0$$

$$0 = \alpha_1 + \alpha_2 \Rightarrow \alpha_1 + \alpha_2 = 0 \rightarrow \textcircled{1}$$

put $n=1$ in eqn (A)

$$f_1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right) \Rightarrow (1+\sqrt{5})\alpha_1 + (1-\sqrt{5})\alpha_2 = 2 \rightarrow \textcircled{2}$$

($\because f_1 = 1$)

solving eqn (1) & (2)

$$x_1 + x_2 = 0 \Rightarrow \boxed{x_1 = -x_2}$$

$$(2) \Rightarrow (1+\sqrt{5})(-x_2) + (1-\sqrt{5})x_2 = 2$$

$$\Rightarrow x_2 [-1-\sqrt{5}+1-\sqrt{5}] = 2$$

$$\Rightarrow x_2 (-2\sqrt{5}) = 2$$

$$\Rightarrow x_2 = \frac{2}{-2\sqrt{5}} = -\frac{1}{\sqrt{5}} \text{ (i.e.) } \boxed{x_2 = -\frac{1}{\sqrt{5}}}$$

$$\text{i.e., } \boxed{x_1 = -x_2 = +\frac{1}{\sqrt{5}}}$$

substituting these values in (A)

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

7. Find the number of positive integers ≤ 1000 and not divisible by 5, 7, 9 and 22

SOLUTION:

Let A denote integers ≤ 1000 that are divisible by 5

Let B denote integers ≤ 1000 that are divisible by 7

Let C denote integers ≤ 1000 that are divisible by 9

Let D denote integers ≤ 1000 that are divisible by 22

$$[A] = \left\lfloor \frac{1000}{5} \right\rfloor = 200, \quad [B] = \left\lfloor \frac{1000}{7} \right\rfloor = 142$$

$$[C] = \left\lfloor \frac{1000}{9} \right\rfloor = 111, [D] = \left\lfloor \frac{1000}{22} \right\rfloor = 45$$

$$[A \cap B] = \left\lfloor \frac{1000}{5 \times 7} \right\rfloor = 28, [A \cap D] = \left\lfloor \frac{1000}{5 \times 22} \right\rfloor = 9$$

$$[A \cap C] = \left\lfloor \frac{1000}{5 \times 9} \right\rfloor = 22, [B \cap C] = \left\lfloor \frac{1000}{7 \times 9} \right\rfloor = 15$$

$$[B \cap D] = \left\lfloor \frac{1000}{7 \times 22} \right\rfloor = 6, [C \cap D] = \left\lfloor \frac{1000}{9 \times 22} \right\rfloor = 5$$

$$[A \cap B \cap C] = \left\lfloor \frac{1000}{5 \times 7 \times 9} \right\rfloor = 3, [A \cap B \cap D] = \left\lfloor \frac{1000}{5 \times 7 \times 22} \right\rfloor = 1$$

$$[A \cap C \cap D] = \left\lfloor \frac{1000}{5 \times 9 \times 22} \right\rfloor = 1, [B \cap C \cap D] = \left\lfloor \frac{1000}{7 \times 9 \times 22} \right\rfloor = 0$$

$$[A \cap B \cap C \cap D] = \left\lfloor \frac{1000}{5 \times 7 \times 9 \times 22} \right\rfloor = 0$$

The no of Integers ≤ 1000 , that are divisible by 5, 7, 9 and 22 is

$[A \cup B \cup C \cup D]$. By principle of inclusion and exclusion.

$$\begin{aligned} [A \cup B \cup C \cup D] &= [A] + [B] + [C] + [D] - [A \cap B] \\ &\quad - [A \cap C] - [A \cap D] - [B \cap C] - [B \cap D] - [C \cap D] \\ &\quad + [A \cap B \cap C] + [A \cap B \cap D] + [A \cap C \cap D] + [B \cap C \cap D] \\ &\quad - [A \cap B \cap C \cap D] \end{aligned}$$

$$\text{i.e., } [A \cup B \cup C \cup D] = (200 + 142 + 111 + 45)$$

$$- (28 + 9 + 22 + 15 + 6 + 5) + (3 + 1 + 1 + 0) - 0$$

$$\text{(i.e.,)} [A \cup B \cup C \cup D] = 418$$

$$\begin{aligned}
 &\therefore \text{The no of integers not divisible} \\
 &\text{by any of } 5, 7, 9 \text{ and } 22 \\
 &= \text{TOTAL} - [A \cup B \cup C \cup D] \\
 &= 1000 - 418 = 582
 \end{aligned}$$

9. (i) prove by mathematical method

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}, \quad n \geq 1.$$

(ii) solve the recurrence relation for $a_n - a_{n-1} - a_{n-2} = 0$ with initial value $a_0 = 1, a_1 = 1$.

SOLUTION:

(i) let $p(n) = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}, \quad n \geq 1$

step 1: For $n=1$

$$p(1) = \frac{1 \cdot (1+1)}{2} = \frac{2}{2} = 1 \Rightarrow 1=1$$

so $p(1)$ is true

step 2: For $n=k$

$$p(k) = 1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

ie, $p(k)$ is true

step 3: For $n=k+1$

$$p(k+1) = (1 + 2 + 3 + \dots + k) + k+1$$

$$\text{i.e., } p(k+1) = \frac{k(k+1)}{2} + (k+1)$$

$$= (k+1) \left(\frac{k}{2} + 1 \right) = (k+1) \frac{(k+2)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

$$p(k+1) = \frac{(k+1)((k+1)+1)}{2}$$

which is $p(k+1)$

That is $p(k+1)$ is true whenever $p(k)$ is true. By the principle of mathematical induction $p(n)$ is true for all positive integer n .

$$(ii) \text{ Given, } a_n - a_{n-1} - a_{n-2} = 0$$

$$\text{let } a_n = r^n \Rightarrow r^n - r^{n-1} - r^{n-2} = 0$$

$$\Rightarrow r^n (1 - r^{-1} - r^{-2}) = 0 \Rightarrow 1 - r^{-1} - r^{-2} = 0$$

$$\Rightarrow 1 - \frac{1}{r} - \frac{1}{r^2} = 0 \Rightarrow \boxed{r^2 - r - 1 = 0, \text{ solving}}$$

$$r = \frac{1 \pm \sqrt{1-4(-1)}}{2 \times 1} = \frac{1 \pm \sqrt{5}}{2}$$

The solution is

$$a_n = \alpha_1 \left(\frac{1-\sqrt{5}}{2} \right)^n + \alpha_2 \left(\frac{1+\sqrt{5}}{2} \right)^n \rightarrow \textcircled{A}$$

To find α_1 and α_2 with $a_0=1$, $a_1=1$
when $n=0$

$$\textcircled{A} \Rightarrow a_0 = \alpha_1 \left(\frac{1-\sqrt{5}}{2} \right)^0 + \alpha_2 \left(\frac{1+\sqrt{5}}{2} \right)^0 \Rightarrow 1 = \alpha_1 + \alpha_2$$

$$\text{(i.e.,)} \quad \alpha_1 + \alpha_2 = 1 \rightarrow \textcircled{1}$$

when $n=1$

$$a_1 = \alpha_1 \left(\frac{1-\sqrt{5}}{2} \right) + \alpha_2 \left(\frac{1+\sqrt{5}}{2} \right)$$

$$1 = \alpha_1 \left(\frac{1-\sqrt{5}}{2} \right) + \alpha_2 \left(\frac{1+\sqrt{5}}{2} \right)$$

$$\Rightarrow \alpha_1 (1-\sqrt{5}) + \alpha_2 (1+\sqrt{5}) = 2 \rightarrow (2)$$

Solving eqn (1) & (2)

From eqn (1) $\Rightarrow \alpha_1 = (1 - \alpha_2)$ in eqn (2)

$$(1 - \alpha_2) (1 - \sqrt{5}) + \alpha_2 (1 + \sqrt{5}) = 2$$

$$1 - \sqrt{5} - \alpha_2 + \alpha_2 \sqrt{5} + \alpha_2 + \alpha_2 \sqrt{5} = 2$$

$$\Rightarrow 2\alpha_2 \sqrt{5} = 2 - 1 + \sqrt{5} = (1 + \sqrt{5})$$

$$\Rightarrow \boxed{\alpha_2 = \frac{(1 + \sqrt{5})}{2\sqrt{5}}}$$

$$(i.e.,) (1) \Rightarrow \alpha_1 = 1 - \alpha_2 = 1 - \left(\frac{1 + \sqrt{5}}{2\sqrt{5}} \right)$$

$$\alpha_1 = \left(\frac{2\sqrt{5} - 1 - \sqrt{5}}{2\sqrt{5}} \right) = \frac{(\sqrt{5} - 1)}{2\sqrt{5}}$$

$$\boxed{\alpha_1 = \frac{\sqrt{5} - 1}{2\sqrt{5}} \text{ (or)} \alpha_1 = - \frac{(1 - \sqrt{5})}{2\sqrt{5}}}$$

Substituting α_1 & α_2 in eqn (A)

$$a_n = - \frac{(1-\sqrt{5})}{2\sqrt{5}} \frac{(1-\sqrt{5})^n}{2^n} + \frac{(1+\sqrt{5})}{2\sqrt{5}} \frac{(1+\sqrt{5})^n}{2^n}$$

$$\boxed{a_n = \frac{(1+\sqrt{5})^{n+1}}{2^{n+1}\sqrt{5}} - \frac{(1-\sqrt{5})^{n+1}}{2^{n+1}\sqrt{5}}}$$

10. (i) prove that $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$ for $n \geq 2$ using by mathematical induction method.

(ii) Find the no of integers ≤ 250 that are not divisible by integers 2, 3 and 5.

SOLUTION:

(i) let $P(n) = \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$, $n \geq 2$

step 1: For $n=2$

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} > \sqrt{2} \Rightarrow 1.702 > 1.414$$

i.e., $P(2)$ is true

step 2: For $n=k$

$$P(k) = \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} > k \text{ is true}$$

step 3: For $n=k+1$

$$P(k+1) = \left(\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} \right) + \frac{1}{\sqrt{k+1}}$$

$$= \sqrt{k} + \frac{1}{\sqrt{k+1}} = \frac{\sqrt{k} \sqrt{k+1} + 1}{\sqrt{k+1}}$$

$$= \frac{\sqrt{k(k+1)} + 1}{\sqrt{k+1}} = \frac{\sqrt{k \cdot k} + 1}{\sqrt{k+1}} = \frac{(k+1)}{\sqrt{k+1}}$$

$P(k+1) = \sqrt{k+1}$ is true. By the principle of M.I $P(n)$ is true for all $n \geq 2$.

(ii) Let A, B and C denote respectively the number of integer between 1 to 250 that are divisible by 2, 3 and 5 respectively

$$[A] = \left\lfloor \frac{250}{2} \right\rfloor = 125, [B] = \left\lfloor \frac{250}{3} \right\rfloor = 83$$

$$[C] = \left\lfloor \frac{250}{5} \right\rfloor = 50, [A \cap B] = \left\lfloor \frac{250}{2 \times 3} \right\rfloor = 41,$$

$$[A \cap C] = \left\lfloor \frac{250}{2 \times 5} \right\rfloor = 25, [B \cap C] = \left\lfloor \frac{250}{3 \times 5} \right\rfloor = 16$$

$$[A \cap B \cap C] = \left\lfloor \frac{250}{2 \times 3 \times 5} \right\rfloor = 8$$

The number of integers between 1 to 250 that are divisible by 2, 3 and 5 by the principle of inclusion and Exclusion

$$[A \cup B \cup C] = [A] + [B] + [C] - [A \cap B] - [A \cap C] - [B \cap C] + [A \cap B \cap C]$$

$$= (125 + 83 + 50) - (41 + 25 + 16) + 8$$

$$= 258 - 82 + 8 = 184$$

The no of integers not divisible by any of 2, 3 and 5 = TOTAL - $[A \cup B \cup C]$
 $= 250 - 184 = 66$

4.(i) From a club consisting of 6 men and 7 women. In how many ways we select a committee of (a) Three men and four women (b) Four person which has at most one man. (c) Four person that has children of both sex.

(ii) There are 6 boys and 5 girls in a group. Find the number of ways 4 people can be selected

- (a) They can be boy (or) girl
 (b) 2 must be boy and 2 must be girl
 (c) They are must be same sex.

SOLUTION:

(i) @ 3 men can be selected from 6 men in 6C_3 ways. 4 women can be selected from 7 women in 7C_4 ways

By product rule, the Committee of 3 men and 4 women can be selected in

$${}^6C_3 \times {}^7C_4 \text{ ways} = \frac{6 \times 5 \times 4}{1 \times 2 \times 3} \times \frac{7 \times 6 \times 5 \times 4}{1 \times 2 \times 3 \times 4}$$

$$= 700 \text{ ways}$$

(b) For the committee of at least one women, we have the following possibilities

- (i) 1 woman and 3 men
 (ii) 2 women and 2 men
 (iii) 3 women and 1 man
 (iv) 4 women and 0 men.

Therefore, the selection can be done in

$$= ({}^7C_1 \times {}^6C_3 + {}^7C_2 \times {}^6C_2 + {}^7C_3 \times {}^6C_1 + {}^7C_4 \times {}^6C_0) \text{ ways}$$

$$= 7 \times 20 + 21 \times 15 + 35 \times 6 + 35 \times 1$$

$$= 140 + 315 + 210 + 35$$

$$= 700 \text{ ways}$$

© For the committee of both sexes we have the following possibilities

- (i) 1 men and 3 women
- (ii) 2 men and 2 women
- (iii) 3 men and 1 women which can be done in

$$= {}^6C_1 \times {}^7C_3 + {}^6C_2 \times {}^7C_2 + {}^6C_3 \times {}^7C_1 \text{ ways}$$

$$= 6 \times 35 + 15 \times 21 + 20 \times 7$$

$$= 210 + 315 + 140 \text{ ways}$$

$$= 665 \text{ ways}$$

(ii) There are 6 boys and 5 girls and we need to select 4 people

① They can be boy or girl (no restriction)

$$\text{Total people} = 6 + 5 = 11$$

$$= {}^{11}C_4 = \frac{11 \times 10 \times 9 \times 8}{1 \times 2 \times 3 \times 4} = 330$$

⑥ 2 must be boys and 2 must be girls
select 2 boys from 6 and 2 girls from 5.

$$= {}^6C_2 \times {}^5C_2 = \frac{6 \times 5}{1 \times 2} \times \frac{5 \times 4}{1 \times 2} = 150$$

⑦ All must be same sex

Either all boys or all girls:

$$= {}^6C_4 + {}^5C_4 = \frac{6 \times 5 \times 4 \times 3}{1 \times 2 \times 3 \times 4} + \frac{5 \times 4 \times 3 \times 2}{1 \times 2 \times 3 \times 4}$$

$$= 15 + 5 = 20$$

6. (i) solve $a_n + 6a_{n-1} + 12a_{n-2} + 8a_{n-3} = 0$
where $a_0 = 1$, $a_1 = 2$, $a_2 = 4$ by characteristic
roots method.

(ii) A committee of 5 to be selected from
6 boys and 5 girls. Find the number
of ways for selecting a committee if it
consists at least one boy and one girl.

SOLUTION:

(i) Given, $a_n + 6a_{n-1} + 12a_{n-2} + 8a_{n-3} = 0$

$$\text{put } a_n = r^n \Rightarrow r^n + 6r^{n-1} + 12r^{n-2} + 8r^{n-3} = 0$$

$$\Rightarrow r^n (1 + 6r^{-1} + 12r^{-2} + 8r^{-3}) = 0$$

$$\Rightarrow r^3 + 6r^2 + 12r + 8 = 0$$

$$\Rightarrow (r+2)^3 = 0 \Rightarrow r = -2, -2, -2$$

The general solution is

$$a_n = (\alpha_1 + \alpha_2 n + \alpha_3 n^2) (-2)^n, n \geq 0$$

$$a_n = (\alpha_1 + \alpha_2 n + \alpha_3 n^2) (-2)^n, \quad n \geq 0 \rightarrow \textcircled{A}$$

To find α_1, α_2 and α_3 with
 $a_0 = 1, \quad a_1 = 2, \quad a_2 = 4$

put $n=0$

$$\textcircled{A} \Rightarrow a_0 = \alpha_1 \Rightarrow \boxed{\alpha_1 = 1}$$

put $n=1$

$$\textcircled{A} \Rightarrow a_1 = (\alpha_1 + \alpha_2 + \alpha_3) (-2)$$

$$\Rightarrow 2 = (\alpha_1 + \alpha_2 + \alpha_3) (-2)$$

$$\Rightarrow \alpha_1 + \alpha_2 + \alpha_3 = -1$$

$$\Rightarrow 1 + \alpha_2 + \alpha_3 = -1$$

$$(\because \alpha_1 = 1)$$

$$\Rightarrow \boxed{\alpha_2 + \alpha_3 = -2} \rightarrow \textcircled{1}$$

put $n=2$

$$\textcircled{A} \Rightarrow a_2 = (\alpha_1 + 2\alpha_2 + 4\alpha_3) 4$$

$$\Rightarrow 4 = (\alpha_1 + 2\alpha_2 + 4\alpha_3) 4$$

$$\Rightarrow \alpha_1 + 2\alpha_2 + 4\alpha_3 = 1$$

$$\Rightarrow 1 + 2\alpha_2 + 4\alpha_3 = 1$$

$$\Rightarrow \boxed{2\alpha_2 + 4\alpha_3 = 0} \rightarrow \textcircled{2}$$

Solving eqn $\textcircled{1}$ & $\textcircled{2}$ $\alpha_2 = -4, \quad \alpha_3 = 2$

$$\therefore a_n = (\alpha_1 + \alpha_2 n + \alpha_3 n^2) (-2)^n$$

$$\boxed{a_n = (1 - 4n + 2n^2) (-2)^n, \quad n \geq 0}$$

(ii)

$$\begin{aligned} & \text{At least one boy and one girl} \\ & = \text{TOTAL ways} - (\text{all boys} + \text{all girls}) \end{aligned}$$

We have 6 boys and 5 girls, and we need a committee of 5 with at least one boy and one girl

total ways to select 5 people from

$$= {}^{11}C_5 = \frac{11 \times 10 \times 9 \times 8 \times 7}{1 \times 2 \times 3 \times 4 \times 5} = 462$$

Subtract cases with no mix

(i.e.,) all boys or all girls

$$= {}^6C_5 + {}^5C_5 = 6 + 1 = 7$$

Required number of ways

$$= \text{Total ways} - (\text{all boys} + \text{all girls})$$

$$= 462 - 7 = 455$$

8. (i) Solve the recurrence relation $S(k) = -3S(k-1) - 3S(k-2) - S(k-3)$ where $S(0) = 0$, $S(1) = -2$, $S(2) = -1$ by characteristic roots method

(ii) A box contains 6 white balls and 5 red balls. Find the number of ways 4 balls can be selected. (a) They can be white or red ball (b) Two must be white and two must be red (c) All must be same colour.

SOLUTION:

(i) The recurrence relation can be rewritten as

$$a_n + 3a_{n-1} + 3a_{n-2} + a_{n-3} = 0.$$

put $\boxed{a_n = \gamma^n}$

$$\Rightarrow \gamma^n + 3\gamma^{n-1} + 3\gamma^{n-2} + \gamma^{n-3} = 0$$

$$\Rightarrow \gamma^n (1 + 3\gamma^{-1} + 3\gamma^{-2} + \gamma^{-3}) = 0$$

$$\Rightarrow 1 + 3\gamma^{-1} + 3\gamma^{-2} + \gamma^{-3} = 0$$

$$\Rightarrow \boxed{\gamma^3 + 3\gamma^2 + 3\gamma + 1 = 0}$$

$$\Rightarrow (\gamma + 1)^3 = 0 \Rightarrow \gamma = -1, -1, -1$$

The general solution is:

$$s(n) = (\alpha_1 + \alpha_2 n + \alpha_3 n^2) (-1)^n, \quad n \geq 0 \quad \text{---} \textcircled{A}$$

To find α_1, α_2 and α_3

by making use of the initial conditions in eqn (A)

Given $s(0) = 0$, $s(1) = -2$ and $s(2) = -1$

put $n=0$

$$\textcircled{A} \Rightarrow s(0) = \alpha_1 \Rightarrow \boxed{\alpha_1 = 0}$$

put $n=1$

$$\textcircled{A} \Rightarrow s(1) = (\alpha_1 + \alpha_2 + \alpha_3) (-1)$$

$$\Rightarrow -2 = (\alpha_1 + \alpha_2 + \alpha_3) (-1)$$

$$\Rightarrow x_1 + x_2 + x_3 = 2$$

$$\Rightarrow x_2 + x_3 = 2 \rightarrow (1) \quad (\because x_1 = 0)$$

put $n=2$

$$(A) \Rightarrow S(2) = (x_1 + 2x_2 + 4x_3) \times 1$$

$$\Rightarrow x_1 + 2x_2 + 4x_3 = -1$$

$$\Rightarrow 2x_2 + 4x_3 = -1 \quad (\because x_1 = 0)$$

$$\hookrightarrow (2)$$

Solving eqn (1) & (2)

$$x_1 = 0, x_2 = \frac{9}{2}; x_3 = (-\frac{5}{2})$$

Substituting these values in (A)

$$S(n) = \left(\frac{9}{2}n - \frac{5}{2}n^2\right) (-1)^n$$

(ii) we have 6 white balls and 5 red balls total = 11 balls

We select 4 balls

(a) They can be white or Red

$$\text{Total ways} = {}^{11}C_4 = 330$$

(b) 2 must be white and 2 must be red. choose 2 white from 6 and 2 red from 5 = ${}^6C_2 \times {}^5C_2$

$$= 15 \times 10 = 150$$

③ All must be same color
(Either all white or all red)

$$= {}^6C_4 + {}^5C_4 = 15 + 5 = 20$$

5. (i) solve the recurrence relation
 $a_n = 8a_{n-1} - 16a_{n-2}$ with initial
condition $a_2 = 16$, $a_3 = 80$.

(ii) solve the recurrence relation
 $a_n + 6a_{n-1} + 12a_{n-2} + 8a_{n-3} = 0$ with
initial condition $a_0 = 1$, $a_1 = 2$, $a_2 = 4$

SOLUTION:

① Given, $a_n - 8a_{n-1} + 16a_{n-2} = 0$

put $a_n = r^n$

$$\Rightarrow r^n - 8r^{n-1} + 16r^{n-2} = 0$$

$$\Rightarrow r^n (1 - 8r^{-1} + 16r^{-2}) = 0$$

$$\Rightarrow 1 - 8r^{-1} + 16r^{-2} = 0$$

$$\Rightarrow r^2 - 8r + 16 = 0$$

$$\text{ie, } r = 4, 4$$

$$\text{ie, } a_n = (\alpha_1 + \alpha_2 n) (4)^n \rightarrow \textcircled{A}$$

to find α_1 and α_2 by making use of
the initial conditions in eqn $\textcircled{A}$

$$a_n = (\alpha_1 + \alpha_2 n) (4)^n \rightarrow \textcircled{A}$$

put $n=2$

$$\textcircled{A} \Rightarrow a_2 = (\alpha_1 + 2\alpha_2) \times 16$$

$$16 = (\alpha_1 + 2\alpha_2) \times 16$$

$$\Rightarrow \alpha_1 + 2\alpha_2 = 1 \rightarrow \textcircled{1}$$

put $n=3$

$$a_3 = (\alpha_1 + \alpha_2 \times 3) \cdot 64$$

$$80 = (\alpha_1 + 3\alpha_2) \cdot 64$$

$$\Rightarrow \alpha_1 + 3\alpha_2 = \frac{5}{4} \rightarrow \textcircled{2}$$

Solving eqn $\textcircled{1}$ & $\textcircled{2}$

$$\alpha_1 = \frac{1}{2}, \alpha_2 = \frac{1}{4}$$

Substituting these values in $\textcircled{A}$

$$a_n = \left(\frac{1}{2} + \frac{1}{4}n\right) (4)^n$$

$\textcircled{ii}$ Refer to B6 (i)

Both problems are the same.

PART-A

1. If 7 colors are used to paint 57 bicycles. How many will have same colors?

Here, No of pigeon = m = No of bicycle = 57

No of Holes = n = No of colours = 7

By generalized pigeon hole principle

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{57-1}{7} \right\rfloor + 1 = 8 + 1 = 9$$

Therefore at least 9 bicycle will have the same colour.

2. How many different words are arranged in the word "INSTITUTION"?

There are 11 letters with 3-I's, 2-N's, 3-T's

$$\therefore \text{Total number of permutations} = \frac{11!}{3! 3! 2! 1! 1! 1!}$$

$$= \frac{11!}{3! 3! 2!} = 554400$$

3. If $a_n = 2^n + 5(3)^n$, Find the first 3 terms

$$a_1 = 2^1 + 5(3)^1 = 2 + 15 = 17$$

$$a_2 = 2^2 + 5(3)^2 = 4 + 45 = 49$$

$$a_3 = 2^3 + 5(3)^3 = 8 + 135 = 143$$

The first 3-terms are 17, 49, 143

4. state inclusion and exclusion principle.

Let x and y be two finite subsets of universal set U . If x and y are disjoint then $|x \cup y| = |x| + |y|$

If x and y are not disjoint then $|x \cup y| = |x| + |y| - |x \cap y|$. This is called the principle of inclusion and Exclusion.

5. state generalised pigeon hole theorem

If m pigeons are assigned to n pigeonholes, then there must be a pigeonhole containing at least $\left\lceil \frac{m-1}{n} \right\rceil + 1$ pigeons, where $\lceil \rceil$ is the greatest integer function

6. How many different words are there in the word "MATHEMATICS"?

There are 11 letters 2-M's, 2-A's, 2-T's,

$$\begin{aligned} \text{Total no of permutations} &= \frac{11!}{2! 2! 2! 1! 1! 1! 1! 1!} \\ &= \frac{11!}{2! \cdot 2! \cdot 2!} = 4989600 \end{aligned}$$

7. How many words will start with same letter among any group of 27 English letters.

Number of objects = $m = 27$ (pigeon)
 Number of categories = $n = 26$ (pigeon-holes)

By the pigeonhole principle

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{27-1}{26} \right\rfloor + 1 = \left\lfloor \frac{26}{26} \right\rfloor + 1 = 1 + 1 = 2$$

(ie.,) At least 2 letters will have the same starting letter.

8. How many different words are arranged in the word "ENGINEERING"?

There are 11 letters 3-E's, 3-N's, 2-G's, 2-I's

$$\text{Total no of permutations} = \frac{11!}{3! 3! 2! 2! 1!}$$

$$= \frac{11!}{3! 3! 2! 2!} = \frac{39916800}{144}$$

$$= 277200$$

9. Find the first two terms of the sequence used by recurrence relations

$$a_n = 6a_{n-1}; \quad a_0 = 2$$

$$a_1 = 6a_0 = 6 \times 2 = 12; \quad a_2 = 6a_1 = 6 \times 12 = 72$$

$\therefore$ Hence the 1st 2 terms are 12 and 72

10. Find ${}^{30}C_{28}$

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using the combination property

$${}^nC_r = {}^nC_{n-r}$$

$$(ie,) {}^{30}C_{28} = {}^{30}C_{30-2} = {}^{30}C_2$$

$$\Rightarrow {}^{30}C_{28} = {}^{30}C_2 = \frac{30 \times 29}{1 \times 2} = 435$$

11. How many permutations are there in the word "MISSISSIPPI"

The word MISSISSIPPI contains 11 letters consisting of 4-I^s, 4-S^s, 2-P^s, M

$$\begin{aligned}\therefore \text{The number of permutations} &= \frac{11!}{4! 4! 2! 1!} \\ &= \frac{11!}{4! 4! 2!} = \frac{3991680}{1152} = 34650\end{aligned}$$

12. Find the permutations in the word "SUCCESS"

Total Letters = 7

Letters consisting 3-S^s, 2-C^s, 1-U, 1-E

No of distinct permutations of a word

$$\text{With repeated letters} = \frac{7!}{3! \cdot 2!} = 420$$

13. How many integers between 1 and 100 that are divisible by 3 and not divisible 7.

Let A and B denote the no between 1 to 100 are divisible by 3 and 7 respectively

$$[A] = \left[\frac{100}{3} \right] = 33, \quad [B] = \left[\frac{100}{7} \right] = 14$$

$$[A \cap B] = \left[\frac{100}{3 \times 7} \right] = 4$$

The number of integers divisible by 3 but not by 7 = $[A] - [A \cap B]$

$$= 33 - 4 = 29$$

14. Find the first two terms of the sequence used by recurrence relation:
 $a_n = 6a_{n-1}, a_0 = 2$

Given $a_n = 6a_{n-1}$

$$a_1 = 6a_0 = 6 \times 2 = 12$$

$$a_2 = 6a_1 = 6 \times 12 = 72$$

Hence the first two terms are: 12 and 72

15. How many positive integers are not exceeding 100 that is divisible by 5.

The numbers divisible by 5 are 5, 10, 15, 20, ..., 100
Total number of numbers $n = \left(\frac{a_n - a_1}{d} \right) + 1 = \frac{100 - 5}{5} + 1$
 $= \frac{95}{5} + 1 = 19 + 1 = 20$